

Solvency Field Theory

The Master Action:
Native Write Rules, Continuum Formulation,
and Observable Readouts
Master Action II Architecture Document

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July 2026

The action is not a Standard Model Lagrangian in disguise. It is the ledger of which settlement writes, curvatures, ports, branches, rigidity costs, quantum supports, and vertices are admissible.

Document status. This document presents the earned architecture of SFT's master action. Every term with an equation has been derived through the V13 gate program. Every term without an equation has a named placeholder with explicit requirements. The architecture is honest: earned terms get equations, open targets get names and constraints, and the boundary between them is visible on every page.

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1 The three-layer architecture

SFT is presented in three layers. Each layer is a coarsening of the previous one. The physics lives in the first. The mathematics lives in the second. The connection to existing physics lives in the third.

Layer 1 — Native write rules. The actual ontology. The native objects are not fields on spacetime. They are write histories, settlement records, and accounting rules. Layer 1 asks: given the current write record, which next writes are legal?

Layer 2 — The SFT continuum action. The mathematical presentation. The continuum action S_{SFT} emerges as the coarse-grained limit of discrete write accounting over many Compton periods. It is SFT's formulation for computing observable predictions. It is not a translation into another framework.

Layer 3 — SM/GR/ Λ CDM correspondence. The translation table connecting each SFT term and readout to its Standard Model, General Relativity, or Λ CDM counterpart. This layer exists for physicists who want to verify that SFT reproduces known results. It is a correspondence, not the native theory.

2 The compact action

$$S_{\text{SFT}} = S_{\text{worldline}} + S_{\text{settlement}} + S_{\text{admissibility}} + S_{\text{frame}} + S_{\text{branch}} + S_{\text{port}} + S_{\text{rigid}} + S_{\text{vertices}}. \quad (1)$$

Eight terms. Each one describes something the universe does. Particles tick. Settlement accumulates. Writes must be legal. Directional holonomy costs energy. Branches flip. Ports complete. History resists change. Interactions conserve protected quantities.

The following eight sections derive each term from its Layer 1 write rule.

3 $S_{\text{worldline}}$ — Worldline histories

WHAT IT IS. The maintained worldline sector. Particles execute Compton ticks at rate $\nu_C = mc^2/h$ along worldlines at total speed c . The write current J_{ops}^μ is the continuum shadow of these discrete ticks.

WHY IT MUST EXIST. Without worldline histories, nothing writes. Without writes, no settlement. Without settlement, no geometry. Without geometry, no universe. The worldline is the source of everything else.

WHAT IT IS MADE OF. The source current density:

$$J_{\text{ops}}^\mu(x) = \sum_i \int d\tau_i \nu_{C,i} u_i^\mu \delta_g^{(4)}(x - X_i(\tau_i)), \quad (2)$$

where $X_i(\tau_i)$ is the i -th particle's worldline, u_i^μ is its four-velocity, and $\nu_{C,i}$ is its Compton frequency. The one-speed budget constrains each worldline:

$$v^2 + u_{\text{int}}^2 = c^2, \quad (3)$$

where v is the translational speed and u_{int} is the internal circulation speed. A particle at rest has $u_{\text{int}} = c$ (pure internal ticking). A photon has $v = c$ and $u_{\text{int}} = 0$ (pure translational propagation).

WHAT IT PRODUCES. The source current J_{ops}^μ feeds every other term. Settlement is sourced by J_{ops}^μ . Admissibility constrains J_{ops}^μ . Forces act on the worldlines that carry J_{ops}^μ . Readouts project from J_{ops}^μ to observables.

WHERE IT WAS EARNED. V12.11 defined J_{ops}^μ . V12.12 established the locally real particle ontology. V12.13 connected closure cadence to mass. V13.0 froze the particle as real energy at c . MA2-G4 reframed the ontology as write records.

WHAT REMAINS OPEN. The exact covariant one-speed constraint C_{1c} in Lagrangian form. The explicit relation between translational and internal speed budgets as a continuum functional.

SM correspondence: matter Lagrangian $\mathcal{L}_m$; stress-energy tensor source.

4 $S_{\text{settlement}}$ — Settlement record

WHAT IT IS. The accumulated write history at each location. How deeply the settlement geometry has been shaped by all past writes. This is the central term of the action — every other term either sources it, constrains it, or reads it.

WHY IT MUST EXIST. Particles write at every Compton tick. Writes are permanent (no-overwrite). The accumulated record of writes IS the geometry. Without this term, writes leave no trace and the geometry doesn't exist.

WHAT IT IS MADE OF. The settlement depth D_{set} sourced by J_{ops}^μ through the settlement accounting kernel K_{set} , with coherence weighting P_{coh} and nonlinear compounding V_{compound} . The weak-field readout must recover the compactness equation:

$$C_A(E, r) = \frac{A_{\text{write}} E}{\hbar c r}, \quad (4)$$

with the gravitational coupling reframed as:

$$G = \frac{A_{\text{write}} c^3}{\hbar}, \quad \gamma = 8\pi^2 = 4\pi \times 2\pi. \quad (5)$$

The central settlement operator candidate is:

$$\boxed{\Delta \ln D_{\text{set}} = P_{\text{coh}} L_{\text{eff}} u - \Phi_{\text{relax}}}, \quad (6)$$

with:

$$L_{\text{eff}} = 3\pi - \frac{1}{2} \ln\left(\frac{25}{13}\right) = 9.097814727066048. \quad (7)$$

The first term is coherence-weighted compounding: how much recurrent writes reinforce the settlement per cycle. The second term is expansion relaxation: how much boundary growth dilutes the settlement per cycle. The equilibrium — where compounding matches relaxation — determines mass.

WHAT IT PRODUCES. Gravity (scalar depth gradient $\rightarrow$ compactness $\rightarrow$ acceleration). Expansion (global write rate over boundary capacity). Mass hierarchy (per-worldline compounding through L_{eff}). Galactic participation (coherence projection through P_{coh}). Quantum support (settlement intensity as write-race participation). Lensing (path-integrated depth).

WHERE IT WAS EARNED. V13.1 earned the compactness equation and G reframing. V13.2 earned the expansion readout. V12.13 earned L_{eff} . Fork 08 earned the participation weighting. MA2-G7C identified Eq. (6) as the minimal toy that passes structural tests.

WHAT REMAINS OPEN. **This is the central open target of SFT.** The exact forms of K_{set} (the settlement accounting kernel), V_{compound} (the nonlinear compounding functional), P_{coh} (the microscopic coherence projector beyond the galactic proxy), Φ_{relax} (the local relaxation operator), and $\mathcal{H}_{\text{settle}}$ (the microscopic settlement-response operator). These are facets of one unsolved problem: the microscopic settlement equation.

SM correspondence: Ricci scalar action $R/(16\pi G)$; metric tensor; stress-energy coupling.

5 $S_{\text{admissibility}}$ — Admissibility constraints

WHAT IT IS. The legal-write gate. Not every mathematical next state is physical. A write is real only if it satisfies all admissibility conditions.

WHY IT MUST EXIST. SFT is not a theory where anything goes. The universe has rules. No-overwrite. One-speed budget. Charge conservation. Port completion. Compactness ceiling. These rules are not external impositions — they are consequences of the settlement ontology.

WHAT IT IS MADE OF. A stack of constraint projectors:

$$\Pi_{\text{NW}} : \quad dD_{\text{set}} \geq 0 \quad (\text{no-overwrite / append-only}), \quad (8)$$

$$\Pi_{1c} : \quad v^2 + u_{\text{int}}^2 = c^2 \quad (\text{one-speed budget}), \quad (9)$$

$$\Pi_{C \leq 1/2} : \quad C_A \leq \frac{1}{2} \quad (\text{compactness ceiling / horizon}), \quad (10)$$

$$\Pi_Q : \quad \partial_\mu J^\mu = 0 \quad (\text{charge conservation}), \quad (11)$$

$$\Pi_{\text{port}} : \quad \text{color-singlet composites only}, \quad (12)$$

$$\Pi_{\text{branch}} : \quad s = \varepsilon_U \eta_C \chi = +1 \quad (\text{weak access gate}). \quad (13)$$

The compactness ceiling $\Pi_{C \leq 1/2}$ deserves emphasis: when local settlement depth reaches $C_A = 1/2$, additional writes do not deepen the well. They extend the saturated region laterally. The horizon area grows. The depth stays at maximum. This is a hard ceiling, not a soft saturation function.

WHAT IT PRODUCES. The arrow of time (append-only). Black hole horizons (compactness ceiling). Conservation laws (Noether currents). Confinement (port completion). Parity violation (branch selection). The second law of thermodynamics (irreversibility of writes).

WHERE IT WAS EARNED. No-overwrite since V1. One-speed since V12.12. Compactness ceiling from V13.1. Charge conservation from R262C. Port completion from R264F. Branch selection from R265F.

WHAT REMAINS OPEN. Formal Lagrangian enforcement of C_{1c} and C_{clear} as explicit constraint functionals.

SM correspondence: unitarity constraints; gauge invariance conditions; horizon theorems.

6 S_{frame} — Directional holonomy (EM)

WHAT IT IS. The energy cost of directional settlement curvature. The electromagnetic sector.

WHY IT MUST EXIST. The particle's worldline tube has a transverse angular coordinate θ_g (R261F). The zero-angle choice is locally arbitrary — relabeling freedom. This forces a compensating connection A_μ . The connection's curvature $F_{\mu\nu}$ is the first relabeling-invariant physical object. Smooth, sign-symmetric, weak-field elasticity forces the cost to be quadratic (R262F).

WHAT IT IS MADE OF.

$$S_{\text{frame}} = \int d^4x \left[-\frac{K_{\text{EM}}}{4} F_{\mu\nu} F^{\mu\nu} + J_{\theta_g}^\mu A_\mu \right], \quad (14)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the directional settlement curvature tensor, $J_{\theta_g}^\mu$ is the conserved directional current from θ_g winding continuity, and K_{EM} is the curvature stiffness connected to α through the two-stage 4π directional projection.

Charge originates from pair production: angular momentum conservation at a symmetric closure vertex forces opposite θ_g holonomies. Opposite windings mesh (attract). Same windings clash (repel). Charge is internal holonomy, invariant under spatial rotation (R261.5F).

WHAT IT PRODUCES. Coulomb $1/r$ potential (R254A). Magnetic fields from moving charges (R262B). Electromagnetic radiation from accelerated charges. Lorentz force. Current conservation $\partial_\mu J^\mu = 0$ (R262C). The coupling framework $\alpha(E) = N_{\text{eff}}^{p(E)}/(16\pi^2)$ (Alpha-137).

WHERE IT WAS EARNED. R262F derived the Maxwell curvature cost from relabeling invariance, smoothness, and sign symmetry. R262B earned full classical dynamics. R262C earned conservation. R261.5F earned charge origin from pair production.

WHAT REMAINS OPEN. The exact value of K_{EM} . The microscopic derivation of α . QED loop corrections.

SM correspondence: $U(1)_{\text{EM}}$ gauge theory; QED.

7 S_{branch} — Branch transitions (Weak)

WHAT IT IS. The mode-transition sector. Within each generation, the charged lepton and neutrino form a doublet. The up-type and down-type quarks form another. The weak interaction is the transition between doublet partners.

WHY IT MUST EXIST. Particles can change which closure mode they occupy. A neutron can become a proton. An up quark can become a down quark. These mode transitions are real physical events that must be accounted for in the action.

WHAT IT IS MADE OF. Non-commuting branch-flip operators producing $SU(2)$ curvature. The $W^\pm$ are charged branch-flip transition resonances. The Z^0 is the neutral mixing mode. The closure-handedness selector $s = \varepsilon_U \eta_C \chi$ restricts weak access to left-handed particles — a fossil of the primordial closure orientation preserved by no-overwrite.

WHAT IT PRODUCES. Beta decay. Neutrino interactions. Parity violation. The electroweak mixing structure.

WHERE IT WAS EARNED. R265F earned the structural kernel: doublet algebra, non-Abelian curvature route, handedness selector.

WHAT REMAINS OPEN. W/Z masses. Weinberg angle. Fermi constant. CKM/PMNS mixing matrices. Decay rates. CP violation.

SM correspondence: $SU(2)_L \times U(1)_Y$ electroweak theory.

8 S_{port} — Port topology (Strong)

WHAT IT IS. The boundary-port completion sector. Quarks carry fractional winding addresses that connect through three boundary ports.

WHY IT MUST EXIST. The boundary protocol rejects incomplete addresses. A single quark's $1/3$ -winding is not a valid external connection. Three quarks bundled (qqq) or a quark-antiquark pair ($q\bar{q}$) complete the address. This topological requirement IS confinement.

WHAT IT IS MADE OF. Three boundary ports from the three-dimensional normal bundle ($D_\perp = 3$) of a worldline in 3+1 spacetime. The $SU(3)$ structural route follows from the permutation symmetry of three ports with eight trace-free connection directions.

WHAT IT PRODUCES. Color charge. Confinement as incomplete-address rejection. Baryons (qqq) and mesons ($q\bar{q}$) as the allowed composites.

WHERE IT WAS EARNED. R264F earned the topology kernel: three ports, three colors, color-neutral bundling, dual-role with generation lanes through commuting normal-bundle readouts.

WHAT REMAINS OPEN. Full $SU(3)$ dynamics. Gluon self-interactions. Asymptotic freedom. The strong coupling α_s . Hadron spectrum.

SM correspondence: $SU(3)_c$ color gauge theory; QCD.

9 S_{rigid} — Frame-rigidity loading

WHAT IT IS. The energy cost of maintaining written frame commitments against the expanding boundary. Every particle's θ_g orientation, branch state, and generation lane were set at formation and must be maintained. Maintenance costs energy. That cost is mass.

WHY IT MUST EXIST. The photon pays nothing — $U(1)$ relabeling does not contradict written history. The W and Z pay heavily — branch-changing rewrites push against the rigidity of already-written settlement. The cost difference IS the mass difference.

WHAT IT IS MADE OF. The microscopic settlement-response operator $\mathcal{H}_{\text{settle}}$, whose readouts include Φ_{relax} (the per-cycle relaxation tax in the settlement equation), the W/Z mass loading, and the 125 GeV rigidity resonance.

WHAT IT PRODUCES. Photon masslessness (relabeling is free). W/Z masses (branch-changing costs energy). The rigidity resonance at 125 GeV. The absolute mass scale through Φ_{relax} . Possibly the running projection $p(E)$ for α .

WHERE IT WAS EARNED. R266F identified the structural correspondence between Φ_{relax} , frame-rigidity loading, and the Standard Model Higgs mechanism. The identification is structural. The operator is not derived.

WHAT REMAINS OPEN. **This is the second major open target.** $\mathcal{H}_{\text{settle}}$ would simultaneously give Φ_{relax} , W/Z masses, rigidity resonance mass, absolute mass scale, and possibly $p(E)$.

SM correspondence: frame-rigidity loading $\leftrightarrow$ Higgs mechanism; rigidity resonance $\leftrightarrow$ Higgs boson (125 GeV); frame-maintenance sector $\leftrightarrow$ Higgs field.

10 S_{vertices} — Interaction vertices

WHAT IT IS. The admissibility rules for common-write events where two or more settlement fields meet and exchange quantum numbers.

WHY IT MUST EXIST. Every scattering, decay, pair production, and annihilation event is a vertex. The vertex rules determine which interactions are allowed and which are forbidden. Without them, the action has dynamics but no interactions.

WHAT IT IS MADE OF.

$$V_{\theta} : \quad \theta_g \text{ winding conserved (charge),} \quad (15)$$

$$V_P : \quad \text{settlement-momentum conserved,} \quad (16)$$

$$V_{\text{branch}} : \quad \text{branch selection by } s = \varepsilon_U \eta_C \chi, \quad (17)$$

$$V_{\text{port}} : \quad \text{all color ports completed,} \quad (18)$$

$$V_{\text{support}} : \quad \text{support-domain overlap required,} \quad (19)$$

$$V_{\text{history}} : \quad \text{no-overwrite preserved across vertex.} \quad (20)$$

A vertex is admissible if and only if all six conditions are satisfied. The product $V_v = V_{\theta} \cdot V_P \cdot V_{\text{branch}} \cdot V_{\text{port}} \cdot V_{\text{support}} \cdot V_{\text{history}}$ must equal 1. If any factor is 0, the interaction is forbidden.

WHAT IT PRODUCES. Conservation laws at every interaction. Selection rules for decays. Forbidden transitions. The Feynman rules in the perturbative expansion (Layer 3 correspondence).

WHERE IT WAS EARNED. R261.5F (pair production vertices). R262C (conservation). R264F (port completion). R265F (branch selection). R260B (ontology definitions).

WHAT REMAINS OPEN. Exact vertex factors and coupling strengths. Scattering cross sections. Decay rates. The full Feynman-rule translation.

SM correspondence: Feynman rules; interaction vertices in QFT.

11 The central settlement equation

The candidate equation that unifies the particle mass program, the galactic participation program, and the expansion accounting:

$$\boxed{\Delta \ln D_{\text{set}} = P_{\text{coh}} L_{\text{eff}} u - \Phi_{\text{relax}}.} \quad (21)$$

Three terms. One equation.

$P_{\text{coh}} L_{\text{eff}} u$ is coherence-weighted compounding. P_{coh} measures how recurrent the source history is (autocorrelation of J_{ops}^μ). L_{eff} is the compounding gain per cycle from the 3:2 sector structure. u is the mode's winding parameter. The product gives the per-cycle settlement reinforcement for sources with coherence P_{coh} .

Φ_{relax} is expansion relaxation. The per-cycle cost of maintaining settlement against boundary growth. Mode-independent. Cancels from mass ratios. Sets the absolute scale when combined with A_{write} .

The equilibrium $P_{\text{coh}} L_{\text{eff}} u = \Phi_{\text{relax}}$ determines mass.

This equation is not yet derived from first principles. It is the minimal functional form that satisfies all known constraints: particle mass ratios, galactic participation, gravity/compactness, black-hole ceiling, and expansion accounting. The microscopic derivation of V_{compound} , P_{coh} , and Φ_{relax} from the Layer 1 write rules is the primary open target of SFT.

12 Observable readouts

The readout map connects S_{SFT} to ten observables. Readouts are projections of the action, not terms in it.

Observable	Projection	Status
Gravity	Scalar depth gradient of $S_{\text{settlement}} \rightarrow C_A \rightarrow a = -c^2 \nabla C_A$	Earned (V13.1)
Expansion	Global write rate from $S_{\text{worldline}}$ over capacity from $S_{\text{admissibility}} \rightarrow H = \dot{N}/I$	Earned structurally (V13.2)
Mass	Per-worldline recurrence of $S_{\text{settlement}} \rightarrow L_{\text{eff}}$ compounding $\rightarrow$ hierarchy	Ratios earned to 0.032%; absolute scale open

Observable	Projection	Status
Participation	Coherence projection of $S_{\text{settlement}} \rightarrow \lambda_{\text{eff}}$	Fork 08: beats fitted λ^* at 0 free parameters
EM	Directional curvature from $S_{\text{frame}} \rightarrow$ Coulomb, Lorentz, radiation	Classical structure earned (R262)
Strong binding	Port topology from $S_{\text{port}} \rightarrow$ confinement	Topology kernel earned (R264F)
Weak transitions	Branch curvature from $S_{\text{branch}} \rightarrow$ decay selection	Structural kernel earned (R265F)
Rigidity masses	Historical loading from $S_{\text{rigid}} \rightarrow$ W/Z /resonance masses	Structural ID (R266F); operator open
Quantum support	Settlement intensity from $S_{\text{settlement}} \rightarrow \psi$, Born rule, Bell	Conditional (V13.5–V13.6)
Lensing	Path integral of $S_{\text{settlement}}$ depth $\rightarrow$ gravitational and expansion lensing	Future gate after normalization

13 SM/GR/ Λ CDM correspondence

SFT term	SM/GR correspon- dence	Translation status
$S_{\text{worldline}}$	Matter source $\mathcal{L}_m$; stress-energy $T_{\mu\nu}$	Structural
$S_{\text{settlement}}$	$R/(16\pi G)$; metric $g_{\mu\nu}$; curvature coupling	Weak-field earned
$S_{\text{admissibility}}$	Unitarity; gauge invariance; horizon theorems	Structural
S_{frame}	$U(1)_{\text{EM}}$ gauge action; QED	Form earned
S_{branch}	$SU(2)_L \times U(1)_Y$; electroweak theory	Structural kernel
S_{port}	$SU(3)_c$ color; QCD	Topology kernel
S_{rigid}	Higgs mechanism; Higgs field; Yukawa couplings	Structural ID
S_{vertices}	Feynman rules; vertex factors	Structural

14 Open targets

All open targets trace to two deep cores.

Core A: The compounding functional. $V_{\text{compound}}, K_{\text{set}}, P_{\text{coh}}, G_{\text{stab}}, C_{1/2}$ saturation. One question: what is the nonlinear settlement compounding law and how does it behave from Compton scale to cosmological scale?

Core B: The relaxation operator. $\Gamma_{\text{settle}}, \Phi_{\text{relax}}, K_{\text{EM}}, \mathcal{H}_{\text{settle}}$. One question: what is the settlement-relaxation operator and how does it set the absolute mass scale, the electromagnetic coupling, and the frame-rigidity masses?

These may be one problem: the compounding and relaxation are the two terms in Eq. (6). Solving the settlement equation from first principles would close both cores simultaneously.

Additional formalism gaps (important but not conceptually deep): exact covariant one-speed constraint C_{1c} ; microscopic no-overwrite gate C_{clear} ; full QCD/electroweak quantitative completion.

15 Normalization watchlist

Two normalization candidates survived testing but are not yet derived:

α projection: $\alpha(E) = N_{\text{eff}}^{p(E)}/(16\pi^2)$. The kernel $1/(16\pi^2)$ from two-stage 4π directional projection is robust. The running exponent $p(E)$ is monotonically increasing with energy ($p_{\text{low}} \approx 0.217, p_Z \approx 0.322$).

Π_λ candidate: $\lambda_{\text{raw}} = 1 + L_{\text{eff}}/(6\pi^2) = 1.1536$. Within 0.2% of the global fitted $\lambda^* \approx 1.156$. Not derived. Multiple geometric decompositions of $6\pi^2$ remain plausible.

Both are on the watchlist: too interesting to discard, not earned enough to freeze.

Final statement

SFT now has a coherent eight-term native action, a central settlement equation candidate, and a sharply constrained operator family. The architecture is earned. The microscopic operator is open. Every sealed result in the V13 program traces to a specific term and readout of S_{SFT} .

The compact action:

$$S_{\text{SFT}} = S_{\text{worldline}} + S_{\text{settlement}} + S_{\text{admissibility}} + S_{\text{frame}} + S_{\text{branch}} + S_{\text{port}} + S_{\text{rigid}} + S_{\text{vertices}}. \quad (22)$$

The central equation:

$$\Delta \ln D_{\text{set}} = P_{\text{coh}} L_{\text{eff}} u - \Phi_{\text{relax}}. \quad (23)$$

Eight terms. One equation. Ten readouts. Architecture earned. Operator open.

Acknowledgments

The MA2 gate program (G0–G8) was developed with ChatGPT/5.5 (OpenAI). The three-layer architecture, the admissibility-functional reframing, the S_{readouts} -is-not-a-term correction, the native write-record ontology correction, the $T_{\text{cont-to-S}}_{\text{SFT}}$ presentation correction, the six-point derivation standard, the $V_{\text{compound}}/_\text{settle}$ gap analysis, and the seven-readout kernel identification were developed with Claude/Opus (Anthropic). The SPARC Fork 08 participation result was developed with Claude/Fable (Anthropic). The V13.0–V13.6 foundation was developed collaboratively with all three.

References

- [1] S. Fosmark, *Solvency Field Theory V13.1: An Information-Theoretic Foundation for Cosmology and Gravity*, Zenodo (2026). <https://zenodo.org/records/20822219>